Project Documentation: Xpandcs Domain Integration

Active Directory Integration Project: Windows Server & Linux Database Server

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Project Team

- **Mostafa Al-Badwaihy**
- **Ali Mansour**
- **Mohamed El-Sherif**

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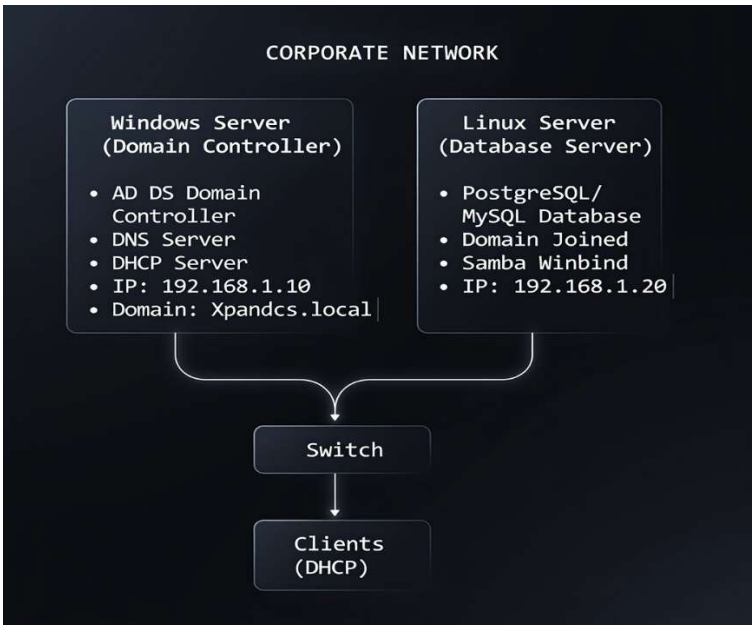
1. Project Overview

This project demonstrates a comprehensive enterprise integration scenario where a Linux server running a database service successfully joins a Windows Active Directory domain. The implementation includes a Windows Server configured with **AD DS**, **DNS**, and **DHCP** roles, alongside a Linux server hosting a **PostgreSQL/MySQL** database that is fully integrated into the Active Directory environment.

Key Objectives

- **Centralized Identity Management**: Enabling domain users to authenticate across heterogeneous systems.
- **Cross-Platform Connectivity**: Ensuring seamless communication between Windows and Linux infrastructure.
- **Infrastructure Automation**: Deploying monitoring and integration scripts to maintain system health.

Network Architecture

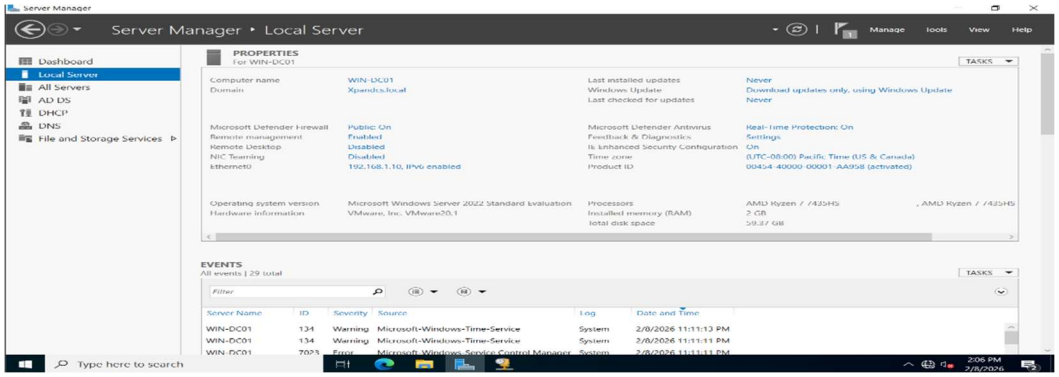


2. Phase 1: Windows Server Infrastructure (WIN-DC01)

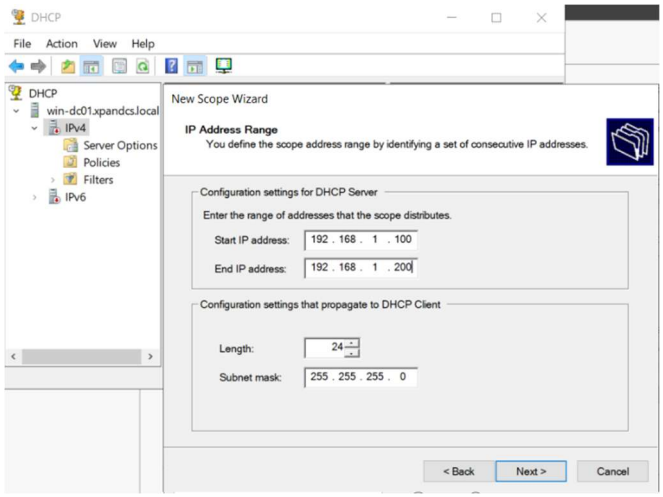
The core infrastructure was built on a Windows Server 2022 Standard Evaluation instance.

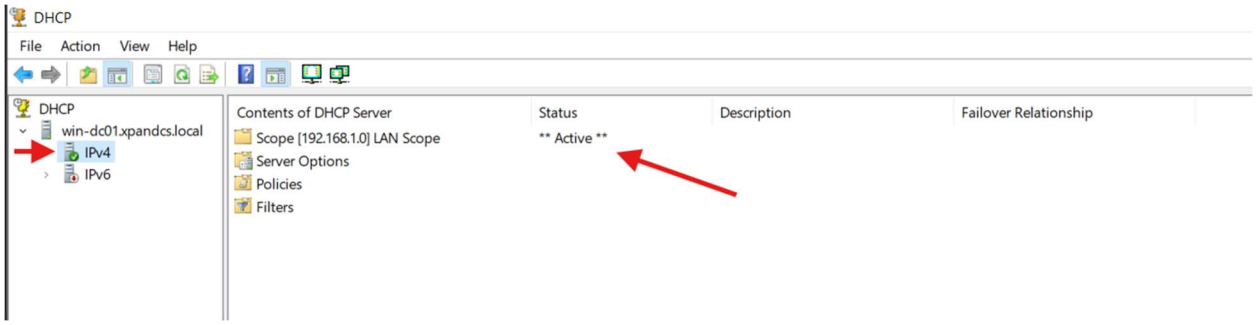
2.1 Services & Roles Installation

- **Active Directory Domain Services (AD DS):** Promoted the server to a Domain Controller for the forest Xpandcs.local.
- **DNS Server:** Installed alongside AD DS to manage internal name resolution for the domain.



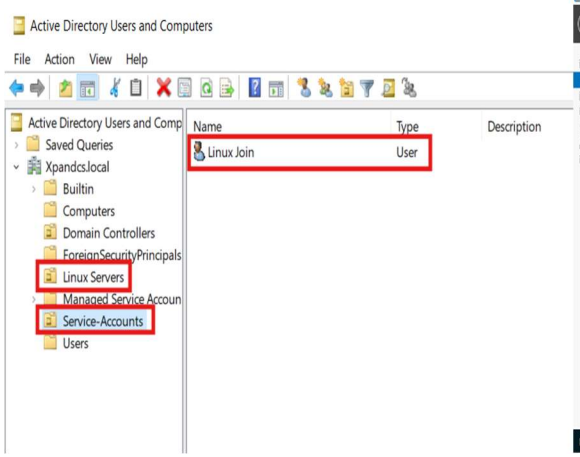
- **DHCP Server:** Authorized and configured with an IPv4 scope.
    - **Scope Range:** 192.168.1.100 to 192.168.1.200.
    - **Subnet Mask:** 255.255.255.0.

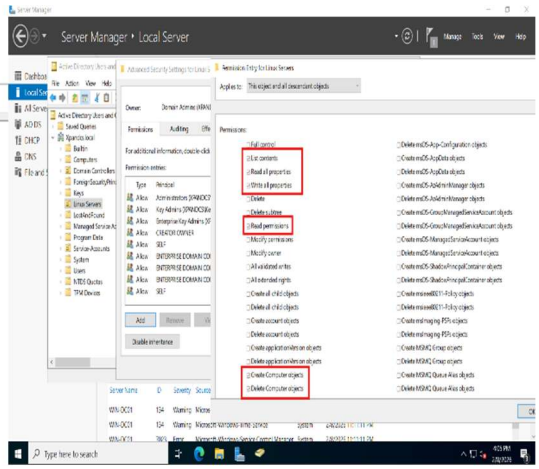




2.2 Active Directory Preparation

- **OU Hierarchy:** Created the Linux Servers and Service-Accounts Organizational Units.
- **Service Account:** Created the Linux Join user specifically for domain join operations.
- **Permission Delegation:** Granted the Linux Join account rights to create and delete computer objects within the Linux Servers OU.



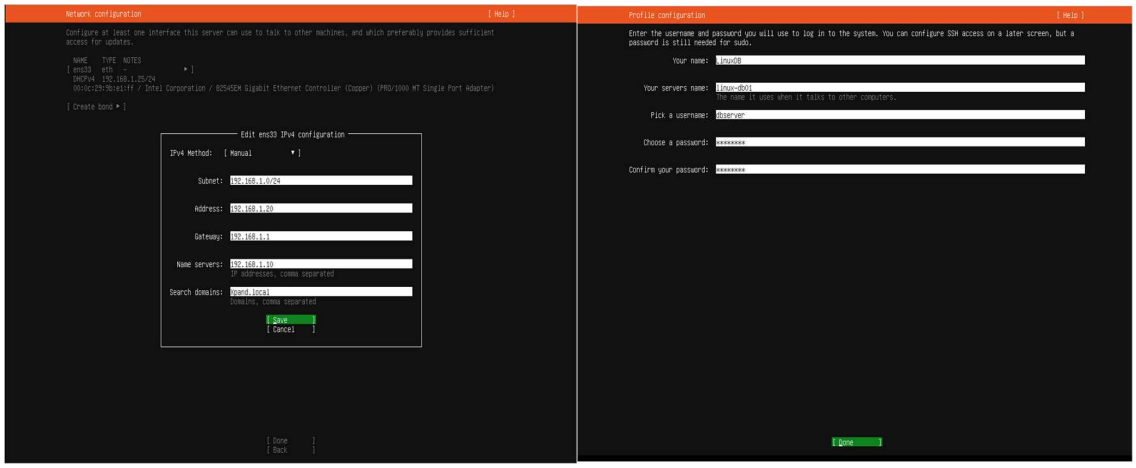


3. Phase 2: Linux Database Server (linux-db01)

The Linux environment was prepared as the secondary node to host the data layer.

3.1 OS & Network Configuration

- **Networking:** Configured the ens33 interface with a manual IP of 192.168.1.20.
- **DNS Alignment:** Set 192.168.1.10 (the DC) as the primary Name Server and Xpandcs.local as the search domain.



3.2 PostgreSQL Deployment

- **Database Creation:** Initialized the companydb instance.
- **Role Management:** Created a local testuser with **Superuser** privileges to manage initial database setup.

```
dbserver@linux-db01: ~
postgres=# \l
                                                  List of databases
   Name    |  Owner   | Encoding | Locale Provider |   Collate   |    Ctype    | ICU Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+-------------+-------------+------------+-----------+-----------------------
 companydb | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =Tc/postgres         +
           |          |          |                 |             |             |            |           | postgres=CTc/postgres+
           |          |          |                 |             |             |            |           | testuser=CTc/postgres
 postgres  | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 template0 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres          +
           |          |          |                 |             |             |            |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres          +
           |          |          |                 |             |             |            |           | postgres=CTc/postgres
(4 rows)

postgres=# |
```

```
postgres=# \du
                                   List of roles
          Role name           |                         Attributes
------------------------------+------------------------------------------------------------
 postgres                     | Superuser, Create role, Create DB, Replication, Bypass RLS
 testuser                     | Superuser
```

4. Phase 3: Domain Integration & Security

The integration was achieved using the realmd and sssd framework.

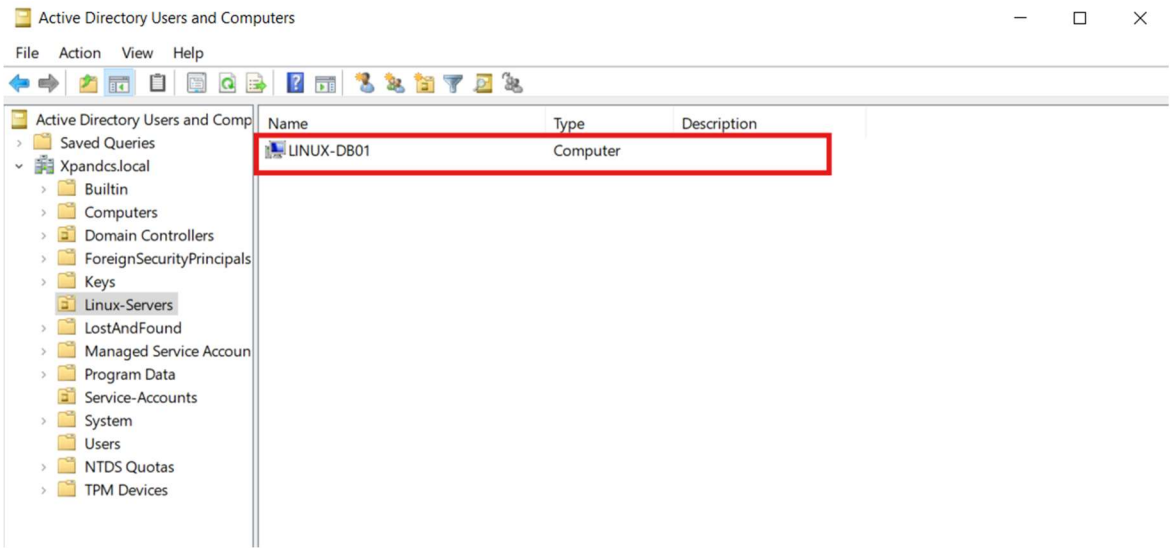
- **Discovery:** Used realm discover Xpandcs.local to locate the Kerberos realm and verify required packages like sssd-tools and adcli.

```
dbserver@linux-db01: /etc/ne
dbserver@linux-db01:/etc/netplan$ realm discover Xpandcs.local
Xpandcs.local
  type: kerberos
  realm-name: XPANDCS.LOCAL
  domain-name: Xpandcs.local
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
xpandcs.local
  type: kerberos
  realm-name: XPANDCS.LOCAL
  domain-name: xpandcs.local
  configured: no
```

- **Authentication:** Configured the system to allow Active Directory users to authenticate using their domain credentials.

```
dbserver@linux-db01:/etc/netplan$ sudo realm join --verbose --computer-ou="OU=Linux-Servers,DC=Xpandcs,DC=local" -U linux Xpandcs.local
 * Resolving: _ldap._tcp.xpandcs.local
 * Performing LDAP DSE lookup on: 192.168.1.10
 * Successfully discovered: Xpandcs.local
Password for linux:
 * Unconditionally checking packages
 * Resolving required packages
 * LANG=C /usr/sbin/adcli join --verbose --domain Xpandcs.local --domain-realm XPANDCS.LOCAL --domain-controller 192.168.1.10 --computer-ou OU=Linux-Servers,DC=Xpandcs,DC=local --login-typ
e user --login-user linux --stdin-password
 * Using domain name: Xpandcs.local
 * Calculated computer account name from fqdn: LINUX-DB01
 * Using domain realm: Xpandcs.local
 * Sending NetLogon ping to domain controller: 192.168.1.10
 * Received NetLogon info from: WIN-DC01.Xpandcs.local
 * Wrote out krb5.conf snippet to /var/cache/realmd/adcli-krb5-D0xTr0/krb5.d/adcli-krb5-conf-FF6OvX
 * Authenticated as user: linux@XPANDCS.LOCAL
 * Using GSS-SPNEGO for SASL bind
 * Looked up short domain name: XPANDCS
 * Looked up domain SID: S-1-5-21-3277612288-229005100-634025399
 * Received NetLogon info from: WIN-DC01.Xpandcs.local
 * Using fully qualified name: linux-db01
 * Using domain name: Xpandcs.local
 * Using computer account name: LINUX-DB01
 * Using domain realm: Xpandcs.local
 * Calculated computer account name from fqdn: LINUX-DB01
 * Generated 120 character computer password
 * Using keytab: FILE:/etc/krb5.keytab
 * A computer account for LINUX-DB01$ does not exist
 ! Couldn't find a computer container in the ou, creating computer account directly in: OU=Linux-Servers,DC=Xpandcs,DC=local
 * Calculated computer account: CN=LINUX-DB01,OU=Linux-Servers,DC=Xpandcs,DC=local
 * Encryption type [3] not permitted.
 * Encryption type [1] not permitted.
 * Created computer account: CN=LINUX-DB01,OU=Linux-Servers,DC=Xpandcs,DC=local
 * Trying to set computer password with Kerberos
 * Set computer password
 * Retrieved kvno '2' for computer account in directory: CN=LINUX-DB01,OU=Linux-Servers,DC=Xpandcs,DC=local
 * Checking RestrictedKrbHost/LINUX-DB01
 *    Added RestrictedKrbHost/LINUX-DB01
 * Checking host/LINUX-DB01
 *    Added host/LINUX-DB01
 * Discovered which keytab salt to use
 * Added the entries to the keytab: LINUX-DB01$@XPANDCS.LOCAL: FILE:/etc/krb5.keytab
 * Added the entries to the keytab: host/LINUX-DB01@XPANDCS.LOCAL: FILE:/etc/krb5.keytab
 * Added the entries to the keytab: RestrictedKrbHost/LINUX-DB01@XPANDCS.LOCAL: FILE:/etc/krb5.keytab
 * /usr/sbin/update-rc.d sssd enable
 * /usr/sbin/service sssd restart
 * Successfully enrolled machine in realm
```

- **PostgreSQL Role Mapping:** Created the Administrator@XPANDCS.LOCAL role in PostgreSQL to permit domain-level database administration.



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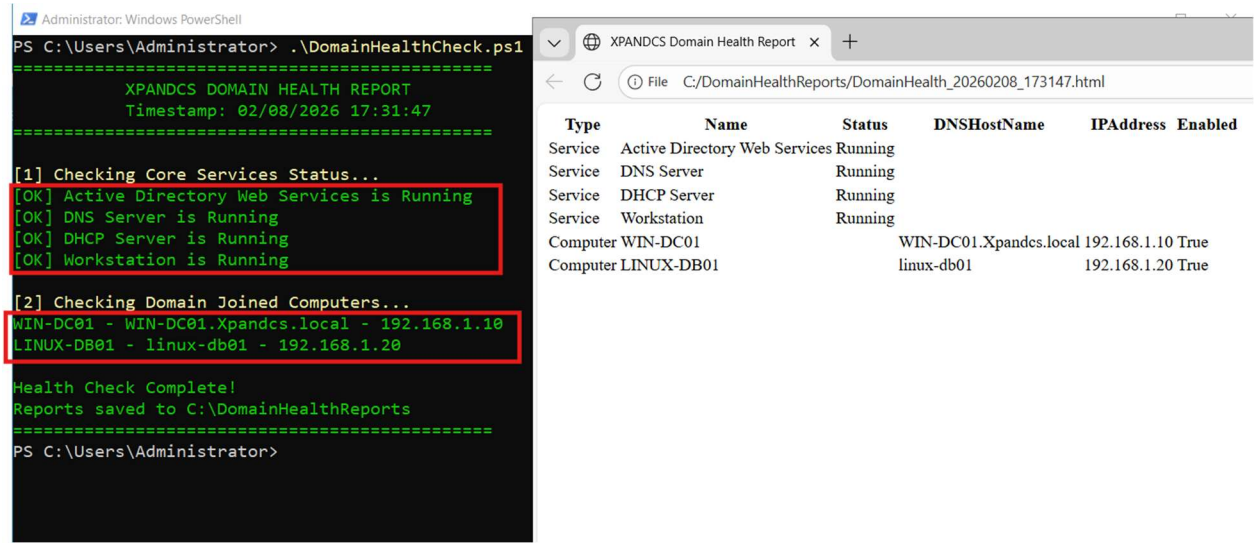
5. Verification & Automation (Script Cases)

To ensure long-term stability, three distinct automation scripts were deployed:

1. **PowerShell (DomainHealthCheck.ps1):** Monitors the health of AD DS, DNS, and DHCP services on the DC.



2. **Bash (ad-integration-check.sh):** A color-coded script that validates DNS, Realm status, Kerberos tickets, and SSSD health on the Linux side.

```
dbserver@linux-db01:~$ bash ad-integration-check.sh
-----------------------------------------------
Checking AD Integration for: Xpandcs.local
-----------------------------------------------
1. DNS Resolution: [ OK ]
2. Domain Join Status: [ OK ]
3. Kerberos Ticket: [ OK ]
4. SSSD Service: [ OK ]
-----------------------------------------------
dbserver@linux-db01:~$
```

3. **Python (db_ad_test.py):** An integration test that connects to companydb and verifies data persistence by writing integration logs.

```
dbserver@linux-db01:~$ python3 db_ad_test.py
--- Xpandcs Domain Integration: Database Test ---
[ OK ] Record created successfully in 'companydb'
[ OK ] Latest Database Entry: (2, 'linux-db01.Xpandcs.local', 'AD_INTEGRATION_SUCCESS', datetime.datetime(2026, 2, 10, 13, 1, 17, 957914))
-----------------------------------------------
Database connection closed.
dbserver@linux-db01:~$ sudo -u postgres psql -d companydb
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1))
Type "help" for help.

companydb=# SELECT * FROM ad_integration_logs;
 id |        test_user         |         status         |         timestamp
----+--------------------------+------------------------+----------------------------
  1 | linux-db01.Xpandcs.local | AD_INTEGRATION_SUCCESS | 2026-02-10 12:57:37.517634
  2 | linux-db01.Xpandcs.local | AD_INTEGRATION_SUCCESS | 2026-02-10 13:01:17.957914
(2 rows)

companydb=#
```

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6. Project Success Criteria

- Linux server successfully joined to Xpandcs Active Directory domain

- Database services running on Linux server

```
dbserver@linux-db01: ~
dbserver@linux-db01:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
     Active: active (exited) since Mon 2026-02-09 14:28:42 UTC; 1h 52min ago
   Main PID: 1243 (code=exited, status=0/SUCCESS)
        CPU: 2ms

Feb 09 14:28:42 linux-db01 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Feb 09 14:28:42 linux-db01 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
dbserver@linux-db01:~$ pg_lsclusters
Ver Cluster Port Status Owner    Data directory              Log file
16  main    5432 online postgres /var/lib/postgresql/16/main /var/log/postgresql/postgresql-16-main.log
dbserver@linux-db01:~$
```

- Xpandcs AD users can authenticate to Linux server

```
dbserver@linux-db01:~$ getent passwd Administrator@Xpandcs.local
administrator@Xpandcs.local:*:1164600500:1164600513:Administrator:/home/administrator@Xpandcs.local:/bin/bash
```

```
dbserver@linux-db01: ~
dbserver@linux-db01:~$ id xpandcs\\linux
uid=1164601105(linux@Xpandcs.local) gid=1164600513(domain users@Xpandcs.local) groups=1164600513(domain users@Xpandcs.local)
```

- DNS resolution working between servers

```
dbserver@linux-db01: ~
dbserver@linux-db01:~$ nslookup xpandcs.local
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   xpandcs.local
Address: 192.168.1.10

dbserver@linux-db01:~$ ping WIN-DC01.xpandcs.local
PING WIN-DC01.xpandcs.local (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=128 time=0.224 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=128 time=0.370 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=128 time=0.509 ms
64 bytes from 192.168.1.10: icmp_seq=4 ttl=128 time=0.449 ms
64 bytes from 192.168.1.10: icmp_seq=5 ttl=128 time=0.457 ms
64 bytes from 192.168.1.10: icmp_seq=6 ttl=128 time=0.494 ms
64 bytes from 192.168.1.10: icmp_seq=7 ttl=128 time=0.408 ms
```

- DHCP serving IP addresses to clients

```
Command Prompt
Windows IP Configuration

   Host Name . . . . . . . . . . . . : Client
   Primary Dns Suffix  . . . . . . . : Xpandcs.local
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : Xpandcs.local

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : Xpandcs.local
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-93-87-A2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::f35c:b920:275e:e2d1%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.100(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, February 9, 2026 7:28:01 AM
   Lease Expires . . . . . . . . . . : Tuesday, February 17, 2026 7:28:00 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.1.10
   DHCPv6 IAID . . . . . . . . . . . : 100666409
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-1C-35-ED-00-0C-29-93-87-A2
   DNS Servers . . . . . . . . . . . : 192.168.1.10
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\Client>
```

- All scripts executing without errors

Conclusion

The Xpandcs Domain Integration project successfully achieved its core objective: bridging the gap between Windows and Linux environments. By implementing a centralized identity model, we have created a secure and efficient infrastructure where the WIN-DC01 Domain Controller governs access to the linux-db01 database server. The deployment of automated monitoring scripts ensures that the integration remains stable, providing a scalable foundation for future enterprise growth within the Xpandcs.local domain.